MATHS MANTHAN
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Subject: Mathematics **CBSE CLASS 10 – PERIODIC ASSESSMENT 07** **Time:** 60 Minutes
Class: Class X (CBSE) **Chapter 11: Areas Related to Circles** **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided where indicated.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer / HOTS (LA) questions of 5 marks each.
6. **Section E** comprises 1 Applied Case-Based question of 4 marks with internal choices.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. A circular garden of radius 21 m is surrounded by a path 3.5 m wide. The area of the path is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 500.5 m^2 (B) 462 m^2 (C) 539 m^2 (D) 484 m^2

Q2. The perimeter of a quadrant of a circle of radius 14 cm is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 22 cm (B) 50 cm (C) 36 cm (D) 44 cm

Q3. If the perimeter of an equilateral triangle is 18 cm, the radius of its circumscribed circle is: [1]

(A) $2\sqrt{3}$ cm (B) $\sqrt{3}$ cm (C) $3\sqrt{3}$ cm (D) 6 cm

Q4. If a sector of a circle of radius 6 cm has perimeter 20 cm, then its area is: [1]

(A) 24 cm^2 (B) 48 cm^2 (C) 12 cm^2 (D) 36 cm^2

Q5. If the difference between the circumference and the radius of a circle is 37 cm, then the diameter of the circle is: $\left(\text{Use } \pi = \frac{22}{7}\right)$ [1]

(A) 7 cm (B) 14 cm (C) 21 cm (D) 28 cm

Q6. **Assertion (A):** Three sectors of central angles $35°, 65°$, and $80°$ cut from a circle of radius r together have an area equal to half the area of the circle.
Reason (R): The sum of the central angles is $35° + 65° + 80° = 180°$, which is half of the total central angle $360°$. [1]

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. The inner circumference of a circular track is 220 m. The track is 7 m wide everywhere. Find the outer radius and outer circumference of the track. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

Q8. If the perimeter of a semicircular protractor is 36 cm, find its diameter. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. A square park has side 50 m. At each of its four corners, a quadrant flower bed of radius 7 m is developed and a circular fountain of diameter 14 m is built at the centre. Find the remaining area of the park. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [3]

Q10. A chord AB of a circle of radius 14 cm subtends an angle of 60° at the centre. Calculate the area of the minor segment. $\left(\text{Use } \pi = \frac{22}{7}, \sqrt{3} = 1.732\right)$ [3]

SECTION D: Long Answer / HOTS Type (2 × 5 = 10 Marks)

Q11. A round table cover has six equal designs. If the radius of the cover is 28 cm:

(a) Prove that each design is a minor segment of a circle of central angle 60° and write the formula for its area.

(b) Calculate the total area of the six designs. $\left(\text{Use } \sqrt{3} = 1.7, \pi = \frac{22}{7}\right)$

(c) Find the total cost of making the designs at the rate of 0.35 per cm^2.

[5]

Q12. An athletic running track consists of two parallel straightaways and two semicircular ends. The length of each straight segment is 106 m, the distance between the two inner parallel lines is 60 m, and the track is 10 m wide everywhere.

(a) Find the inner perimeter of the track.

(b) Find the outer perimeter of the track.

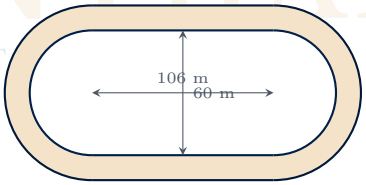
(c) Calculate the total surface area of the running track. $\left(\text{Use } \pi = \frac{22}{7}\right)$

[5]

SECTION E: Case-Based / Applied Situation (1 × 4 = 4 Marks)

Q13. Athletic Stadium Running Track Engineering

A sports stadium features an international regulation running track comprising two parallel straight sections of 106 m each, connected by two semicircular ends. The distance between the inner parallel tracks is 60 m, and the track has a uniform width of 10 m.



(i) Find the radius of the inner semicircular boundary of the track. [1]

(ii) Calculate the total distance a runner covers in one full lap along the inner edge. [1]

(iii) Calculate the total surface area of the track, OR find the total distance covered in one lap along the outer edge. $\left(\text{Use } \pi = \frac{22}{7}\right)$ [2]

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